

Week 04. Cause and Effect in the RCT. Hypothesis testing. T-tests. Exmples. Central Limit Theorem.

Part 1: Statistics and Econometrics. MK68-2024/25

Eduard Bukin

eduard.bukin@agrار.uni-giessen.de

Justus Liebig University Giessen

Institute of Agricultural Policy and Market Research

Content

- Hypothesis testing
- HT Examples
- Examples of Hypothesis Testing
- Effects of a de-worming pill
- Example: Conditional Cash Transfers (CCTs)
- Homework
- References

Hypothesis testing

A statistical hypothesis test is a **method of statistical inference** used to decide **whether the data at hand sufficiently supports a particular hypothesis**.

Hypothesis testing allows us to make probabilistic statements about population parameters.

- Read more in Appendix C-6 in ([Wooldridge 2020](#)).

Hypothesis testing

What?

- Hypothesis Test: A statistical test of the **null hypothesis** against an **alternative hypothesis** ([Wooldridge 2020](#)).

Why?

- We never know the true parameters of the population.
- We must infer about the population from samples.
- HT establishes certainty behind our inference (probability or an error).

How?

- Following Neyman-Pearson's ([Neyman and Pearson 1933](#)) approach to hypothesis testing.

Neyman-Pearson's approach

- Assumes that an **infinite number of repeated samples** can be drawn from the same population.
- Aims to **minimize the risk of making false conclusions** about the relationship.
- Controls the probability of error when stating, “**we reject the null hypothesis.**”

Hypothesis Testing: Steps

1. H_0 : Null/Zero Hypothesis;
 - $H_0 : \mu = 0$
2. H_1 : Alternative Hypothesis;
 - $H_1 : \mu \neq 0$
3. **Estimate parameters:**
 - $\hat{\mu} = \widehat{\text{mean}}$, and
 - $s = \widehat{SD}$
4. **Compute the test statistics;**
 - usually t statistics
5. **Conclude:**
 - to reject $H_0 = \text{accept } H_1$, or
 - fail to reject $H_0 = \text{accept } H_0$
6. **Calculate probabilities of making an error.**
 - probability value (aka “p-value”)

Hypothesis Testing may result in two errors

Type I error

- false positive
- probability value: “p-value”

The error of rejecting H_0 , when H_0 is the **TRUE** hypothesis

Type II error

- false negative

The error of accepting H_0 , when H_0 is the **FALSE** hypothesis

⚠ NEVER KNOWN to the researchers!

p-value probability value

Neyman-Pearson's approach to HT suggests:

- reject H_0 , when
- p-value (probability of the **Type I error**)
- is below **the level of significance α** .

That means:

- when we reject H_0 , the probability that we've made a mistake is below α

Thus, **we reject H_0 only and only if p-value $< \alpha$** .

α is the level of significance

- Maximum probability of a Type I error in hypothesis testing.
- In social sciences, the level of significance
 - should not be higher than 5% ($\alpha = 0.05$).
 - $\alpha = 0.01$, $\alpha = 0.001$ are allowed.

HT Examples

Example (1): Stunting and education

Can educating and training parents about nutrition reduce stunting in children?

Our data measures the change in stunting levels before and after training for 500 individuals.

- Formulate the hypotheses: H_0 and H_1 .
- Provide examples of Type I and Type II errors in the context of your hypothesis.

Hypotheses examples

Null hypothesis:

- $H_0 : \Delta_{\text{stunting}} = 0$

Alternative hypothesis:

- $H_1 : \Delta_{\text{stunting}} \neq 0$

- $H_1 : \Delta_{\text{stunting}} > 0$

- $H_1 : \Delta_{\text{stunting}} < 0$

Errors examples

Type I error:

- 1/4 We conclude **to reject** $H_0: \Delta_{\text{stunting}} \neq 0$,
- 2/4 But this conclusion is driven **by a random variation in the sample**.
- 3/4 In the population, training had nothing to do with stunting.
- 4/4 Repeated sampling will probably fail to show a difference.

Type II error:

- 1/3 We conclude **to accept** H_0 : stunting rate did not change after training.
- 2/3 In fact, the Training helped, but...
- 3/3 other factors led to the food shortage in the survey year.

Example (2)

Question: What % of ravens are White?

Hypothesis:

- H_0 : White ravens make up 0% of the population
- H_1 : Share of white raven is not zero ($> 0\%$)

Data:

- we went to a park;
- we saw 100 ravens (or crows);
- **no white ravens!**

Results: the share of white ravens is 0!

Conclusion: **fail to reject H_0 (Accept H_0)**

Example (2): Raven color

Conclusion: Accept H_0 (there are no white ravens)

Did we make any errors? If yes, which one?

- We've made the **type II error: false negative**.

Because:

- We do not know if H_0 is a true hypothesis.
- We did not use theory (did not study ornithology).
- **The theory has to justify our choice of the H_0 .**



Homework on hypothesis testing

Be prepared to answer the following questions in response to a hypothetical research question prompt during an exam.

1. What is the outcome variable and what is the treatment? Describe how you would measure those.
2. Define the factual and counterfactual outcomes.
3. Formulate H_0 and H_1 .
4. What plausible causal channel(s) runs directly from the treatment to the outcome?
5. Discuss Type I and Type II errors in the context of your hypothesis.
6. Provide examples of factors that could lead to these errors.
7. Propose an RCT experiment that could help answer these questions.
8. Explain what makes an RCT a good/bad solution to the research question.

Research questions:

- How does students' performance in a class vary across different years of enrollment?
- Are yields higher on farms that use precision agriculture?
- How effective are cash disbursements in alleviating poverty?
- What is the poverty-reducing effect of schooling?
- Many firms, particularly in southern European countries, are small, and owned and run by families. Are family owned firms growing more slowly than firms with a dispersed ownership?
- What is the effect of studying economics rather than sociology on the salaries of university graduates?
- What is the effect of mortgage interest rates on the number of new housing starts?

Examples of Hypothesis Testing

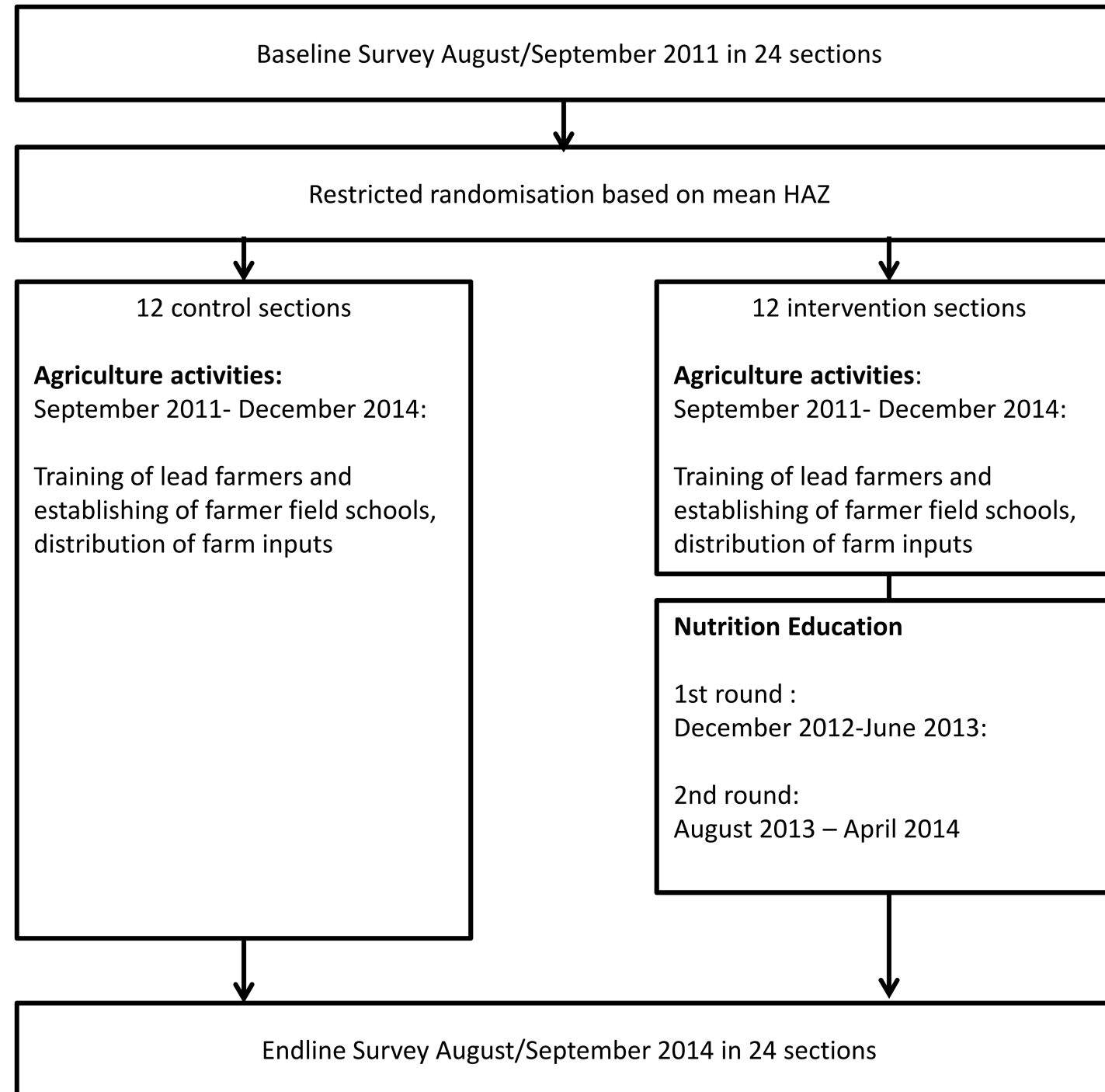
The paper ([Kuchenbecker et al. 2017](#)) aimed to assess the potential of community-based nutrition education to improve height-for-age z-scores in children aged 6 to 23 months.

- The authors conducted a randomized controlled trial (RCT) in Malawi over 5 years.
- Data were collected for both the control and treatment groups at baseline and at the end line.

Kuchenbecker, Judith, ... and Irmgard Jordan. 2017. "Nutrition Education Improves Dietary Diversity of Children 6-23 Months at Community-Level: Results from a Cluster Randomized Controlled Trial in Malawi." Edited by Jacobus P. van Wouwe. PLOS ONE 12 (4): e0175216.

<https://doi.org/10.1371/journal.pone.0175216>

Kuchenbecker et al. (2017): Research design



Kuchenbecker et al. (2017)

- If the paper finds a difference in nutrition statuses between the control and treatment groups, does it mean that education has a causal effect on nutrition?
 - Yes! But...
 - Only if the treatment and control groups are similar on average in (all) other factors.
 - Such factors include income, education, household characteristics, skills, agricultural abilities, and so on...

Checks and balances

To identify that factors (age, education, ...) are **NOT DIFFERENT** between two groups, we need to:

- Perform statistical hypothesis testing.
- Formulate the hypotheses:
 - H_0 : Factor A in the treatment group is equal to that in the control group.
 - H_1 : Factor A in the treatment group is NOT equal to that in the control group.
- Conclude:
 - We fail to reject H_0 .
 - *This means there is no difference between the control and treatment groups.*

Checks and balances: the table

This table is essential for evaluating whether the treatment and control groups of the RCT are similar on average.

The table usually shows:

- The estimated mean and standard deviation (SD) in the control and treatment groups.
- The statistical test for the difference in means.
- The probability value of the Type I error (p-value).

Kuchenbecker et al. (2017): Checks and balances tables 2-3

	Baseline 2011 (n = 832)					Endline 2014 (n = 959)				
	Control		Intervention			Control		Intervention		
	n	mean ($\pm$ SD)	n	mean ($\pm$ SD)	p^+	n	mean ($\pm$ SD)	n	mean ($\pm$ SD)	p^+
Household size	412	5.8 (2.3)	419	5.7 (2.1)	0.49	466	5.3 (1.9)	493	5.3 (2.0)	0.78
Wealth status	413	-0.2 (3.4)	418	0.5 (3.7)	0.01	466	-0.3 (3.8)	493	0.1 (3.7)	0.10
Years of education of caregiver	412	5.9 (3.0)	418	6.2 (2.9)	0.12	466	6.5 (2.9)	493	6.8 (2.7)	0.09
Years of education of household head	409	8.1 (2.5)	416	8.1 (2.4)	0.79	449	8.1 (2.7)	480	8.3 (2.7)	0.41

Full [table 2](#) and [table 3](#)

- Formulate H0 and H1 about **each variable** in the table.
- Perform statistical HT about the mean difference between treatment and control groups..
- Conclude if each variable **is the same (is different)** between the treatment and control groups.

Balance of the HH size

	Baseline 2011 (n = 832)					Endline 2014 (n = 959)				
	Control		Intervention			Control		Intervention		
	n	mean ($\pm$ SD)	n	mean ($\pm$ SD)	p^+	n	mean ($\pm$ SD)	n	mean ($\pm$ SD)	p^+
Household size	412	5.8 (2.3)	419	5.7 (2.1)	0.49	466	5.3 (1.9)	493	5.3 (2.0)	0.78
Wealth status	413	-0.2 (3.4)	418	0.5 (3.7)	0.01	466	-0.3 (3.8)	493	0.1 (3.7)	0.10
Years of education of caregiver	412	5.9 (3.0)	418	6.2 (2.9)	0.12	466	6.5 (2.9)	493	6.8 (2.7)	0.09
Years of education of household head	409	8.1 (2.5)	416	8.1 (2.4)	0.79	449	8.1 (2.7)	480	8.3 (2.7)	0.41

- H_0 : Average household size is equal between Control and treatment;
 - $H_0 : \text{HH Size}_{\text{Control}} = \text{HH Size}_{\text{Treatment}}$
- H_0 : Average HH size is unequal
 - $H_1 : \text{HH Size}_{\text{Control}} \neq \text{HH Size}_{\text{Treatment}}$
- Probability of the type I error: 0.49 (p-value)
- Fail to reject H_0 . Because:
- The difference between C and I is insignificant at the 5% level of significance.

Remember

The checks and balances table must indicate that **treatment and control groups are similar (=)** on average.

- What if the treatment and control groups are not similar on average?
 - Our outcome variable may also not be similar.
 - However, this difference may be caused by factors other than the treatment.
 - The true effect of the treatment could still be zero!

How to identify if the intervention helped?

Because Kuchenbecker et al. (2017) is an RCT, we can:

- Compare means of the outcome variable between two groups, by
- Performing a statistical hypothesis testing (same as in checks and balances),
- But now **we want to reject** (find a difference) the H_0 at the significance level α

Kuchenbecker et al. (2017) table 4: Intervention outcomes

	Baseline 2011					Endline 2014						
	C	I		C	I	C	I		C	I	Intervention effect	
	(n = 413)	(n = 419)		(n = 413)	(n = 419)	(n = 466)	(n = 493)		(n = 466)	(n = 493)		
Food groups	prevalences unadjusted (%)		<i>p</i> ⁺	prevalences adjusted* (%)		prevalences unadjusted (%)		<i>p</i> ⁺	prevalences adjusted* (%)		(%)	<i>p</i>
Grains, roots & tubers	96.4	98.3	0.08	96.6	98.2	95.2	97.6	0.06	95.3	97.5	0.66	0.69
Grains	96.4	98.3	0.08	96.7	98.2	94.7	97.4	0.04	94.8	97.3	0.92	0.58
Roots & tubers	24.9	25.1	0.97	26.1	24.7	19.4	18.7	0.79	18.9	18.6	1.13	0.78
Legumes & nuts	57.1	67.5	<0.01	59.4	66.7	65.4	77.9	0.01	64.8	77.3	4.86	0.35
Groundnuts	35.8	46.3	<0.01	37.4	45.7	38.9	62.4	<0.01	38.7	61.9	14.96	<0.01
Other legumes & nuts	41.2	49.4	0.02	43.1	48.6	50.9	61.4	<0.01	50.5	60.9	4.87	0.34
Dairy products	12.9	13.4	0.83	13.5	12.5	15.6	20.9	0.03	16.2	20.6	5.47	0.15
Flesh foods	31.2	35.3	0.21	32.3	34.5	27.4	38.2	<0.01	27.0	38.4	9.20	0.09
Meat	15.5	16.7	0.64	16.0	15.9	12.8	18.9	0.01	13.0	19.1	6.18	0.11
Organ meat	2.7	2.8	0.80	2.8	2.3	1.3	2.6	0.14	1.2	2.6	1.96	0.16
Fish & seafood	19.9	23.2	0.25	20.6	22.9	18.5	24.1	0.04	18.2	24.1	3.57	0.43
Eggs	12.1	13.4	0.59	12.8	12.9	6.1	16.3	<0.01	6.1	16.2	9.92	<0.01
Vitamin A rich fruit & vegetables	69.0	75.7	0.03	71.1	75.6	71.6	77.5	0.04	70.2	77.3	2.62	0.53
Vitamin A rich roots & tubers	11.9	7.6	0.04	12.1	7.3	12.2	12.9	0.74	12.1	13.2	5.82	0.11
Vitamin A rich vegetables	65.9	73.0	0.03	68.0	72.9	67.8	73.3	0.07	66.4	73.1	1.77	0.67
Vitamin A rich fruit	10.4	16.2	0.01	10.8	16.1	6.7	16.9	<0.01	6.5	16.9	5.14	0.25
Other fruit & vegetables	72.4	78.3	<0.05	74.6	78.0	73.5	82.5	<0.01	72.5	82.0	6.16	0.11
IYCF indicators												
MDD	56.3	62.8	0.06	59.9	61.9	56.3	71.7	<0.01	55.5	71.1	12.70	0.01
MMF	80.0	89.2	<0.01	80.4	88.6	81.0	90.6	<0.01	81.6	90.3	0.43	0.91
MAD	47.9	57.7	<0.01	55.2	56.7	47.7	66.5	<0.01	47.3	65.8	11.86	0.02

Watch now:

How to Read Economics Research Papers: Randomized Controlled Trials (RCTs)

https://youtu.be/s-_3s30Meqs

See (Carter, Greenberg, and Walker 2017) that underpins the video.

At home:

Practice:

- Formulate H_0 and H_1 .
- Perform statistical hypothesis testing about the mean difference between treatment and control.
- For each variable in
 - Table 4 in Kuchebencker et al. ([2017](#)).
 - Table 1 in P. Gertler ([2004](#))

Effects of a de-worming pill

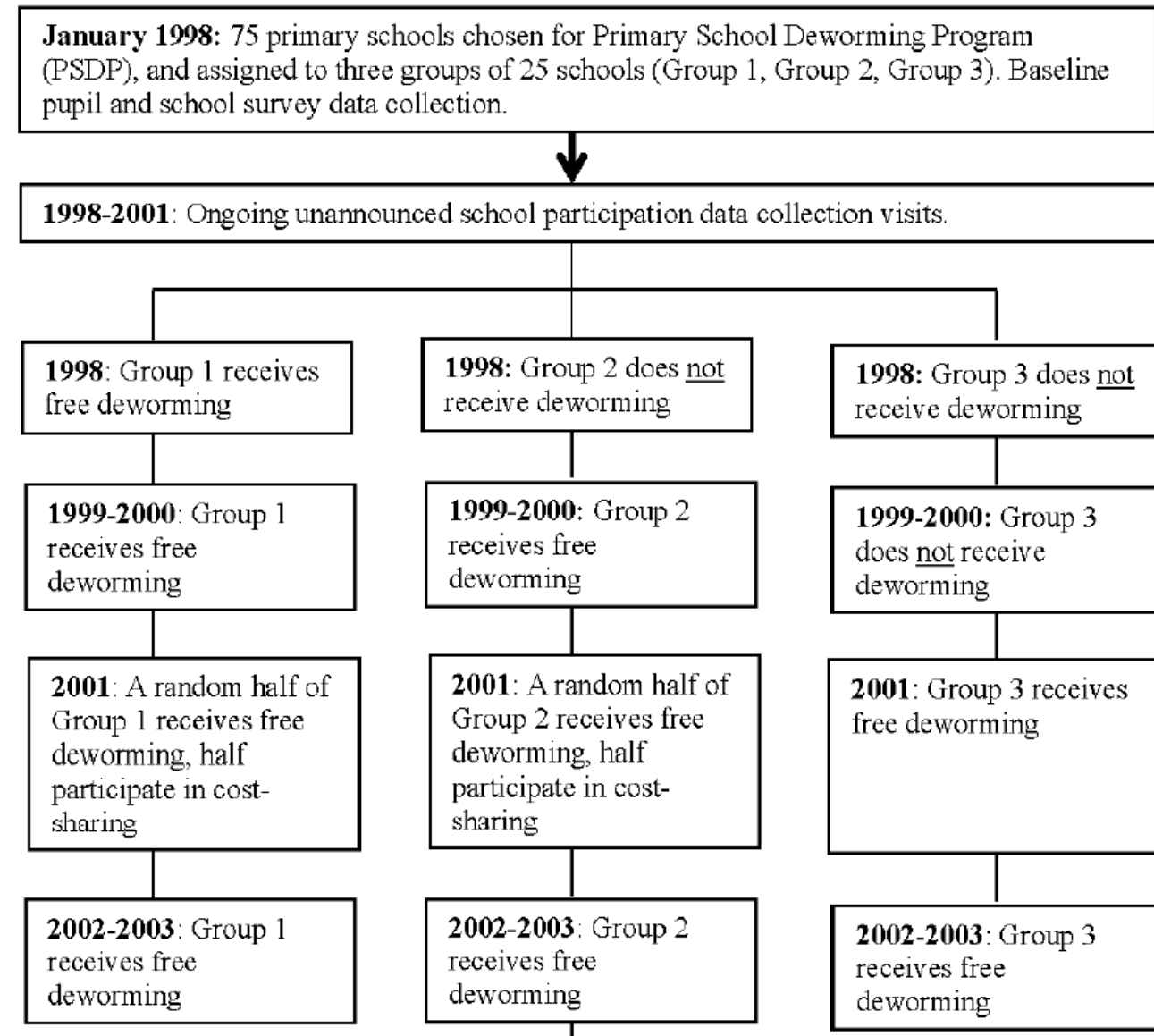
In 1998 an NGO initiated

- the Primary School Deworming Project [PSDP]
- in Busia District (Busia County), western Kenya that is fairly representative of rural Kenya
- in 75 randomly selected schools (32,000 pupils) with a helminths prevalence rate over 90%!

PSDP implementation timeline

Source: ([Hamory et al. 2021](#)), supplementary materials.

Fig. S1. Project Timeline of the Primary School Deworming Program (PSDP) and the Kenya Life Panel Survey (KLPS)



Checks and balances between groups

Source: (Miguel and Kremer 2004) table 1.

- See the full table and interpret it.
- Is there any difference between treatment and control groups?

TABLE I					
1998 AVERAGE PUPIL AND SCHOOL CHARACTERISTICS, PRE-TREATMENT ^a					
	Group 1 (25 schools)	Group 2 (25 schools)	Group 3 (25 schools)	Group 1 – Group 3	Group 2 – Group 3
<i>Panel A: Pre-school to Grade 8</i>					
Male	0.53	0.51	0.52	0.01 (0.02)	–0.01 (0.02)
Proportion girls <13 years, and all boys	0.89	0.89	0.88	0.00 (0.01)	0.01 (0.01)
Grade progression (= Grade – (Age – 6))	–2.1	–1.9	–2.1	–0.0 (0.1)	0.1 (0.1)
Year of birth	1986.2	1986.5	1985.8	0.4** (0.2)	0.8*** (0.2)
<i>Panel B: Grades 3 to 8</i>					
Attendance recorded in school registers (during the four weeks prior to the pupil survey)	0.973	0.963	0.969	0.003 (0.004)	–0.006 (0.004)
Access to latrine at home	0.82	0.81	0.82	0.00 (0.03)	–0.01 (0.03)
Have livestock (cows, goats, pigs, sheep) at home	0.66	0.67	0.66	–0.00 (0.03)	0.01 (0.03)
Weight-for-age Z-score (low scores denote undernutrition)	–1.39	–1.40	–1.44	0.05 (0.05)	0.04 (0.05)
Blood in stool (self-reported)	0.26	0.22	0.19	0.07** (0.03)	0.03 (0.03)
Sick often (self-reported)	0.10	0.10	0.08	0.02** (0.01)	0.02** (0.01)
Malaria/fever in past week (self-reported)	0.37	0.38	0.40	–0.03 (0.03)	–0.02 (0.03)

Immediate effect on health

Source: (Miguel and Kremer 2004) Table 5.

- Check the full table and interpret it.
- What effects were and were not caused by deworming treatment?

TABLE V			
JANUARY TO MARCH 1999, HEALTH AND HEALTH BEHAVIOR DIFFERENCES BETWEEN GROUP 1 (1998 TREATMENT) AND GROUP 2 (1998 COMPARISON) SCHOOLS ^a			
	Group 1	Group 2	Group 1 – Group 2
<i>Panel A: Helminth Infection Rates</i>			
Any moderate-heavy infection, January–March 1998	0.38	–	–
Any moderate-heavy infection, 1999	0.27	0.52	–0.25*** (0.06)
Hookworm moderate-heavy infection, 1999	0.06	0.22	–0.16*** (0.03)
Roundworm moderate-heavy infection, 1999	0.09	0.24	–0.15*** (0.04)
Schistosomiasis moderate-heavy infection, 1999	0.08	0.18	–0.10* (0.06)
Whipworm moderate-heavy infection, 1999	0.13	0.17	–0.04 (0.05)
<i>Panel B: Other Nutritional and Health Outcomes</i>			
Sick in past week (self-reported), 1999	0.41	0.45	–0.04** (0.02)
Sick often (self-reported), 1999	0.12	0.15	–0.03** (0.01)
Height-for-age Z-score, 1999 (low scores denote undernutrition)	–1.13	–1.22	0.09* (0.05)
Weight-for-age Z-score, 1999 (low scores denote undernutrition)	–1.25	–1.25	–0.00 (0.04)
Hemoglobin concentration (g/L), 1999	124.8	123.2	1.6 (1.4)
Proportion anemic (Hb < 100g/L), 1999	0.02	0.04	–0.02** (0.01)

Immediate effect on school participation

Source: (Miguel and Kremer 2004) Table 8.

- Check the full table and interpret it.

TABLE VIII
SCHOOL PARTICIPATION, SCHOOL-LEVEL DATA^a

	Group 1 (25 schools)	Group 2 (25 schools)	Group 3 (25 schools)		
<i>Panel A:</i>					
<i>First year post-treatment (May 1998 to March 1999)</i>	<i>1st Year Treatment</i>	<i>Comparison</i>	<i>Comparison</i>	<i>Group 1 – (Groups 2 & 3)</i>	<i>Group 2 – Group 3</i>
Girls <13 years, and all boys	0.841	0.731	0.767	0.093*** (0.031)	–0.037 (0.036)
Girls ≥13 years	0.864	0.803	0.811	0.057** (0.029)	–0.008 (0.034)
Preschool, Grade 1, Grade 2 in early 1998	0.795	0.688	0.703	0.100*** (0.037)	–0.018 (0.043)
Grade 3, Grade 4, Grade 5 in early 1998	0.880	0.789	0.831	0.070*** (0.024)	–0.043 (0.029)
Grade 6, Grade 7, Grade 8 in early 1998	0.934	0.858	0.892	0.059*** (0.021)	–0.034 (0.026)
Recorded as “dropped out” in early 1998	0.064	0.050	0.030	0.022 (0.018)	0.020 (0.017)
Females ^b	0.855	0.771	0.789	0.076*** (0.027)	–0.018 (0.032)
Males	0.844	0.736	0.780	0.088*** (0.031)	–0.044 (0.037)
<i>Panel B:</i>					

Long-term economic effects (1)

Source: (Hamory et al. 2021) Table 1.

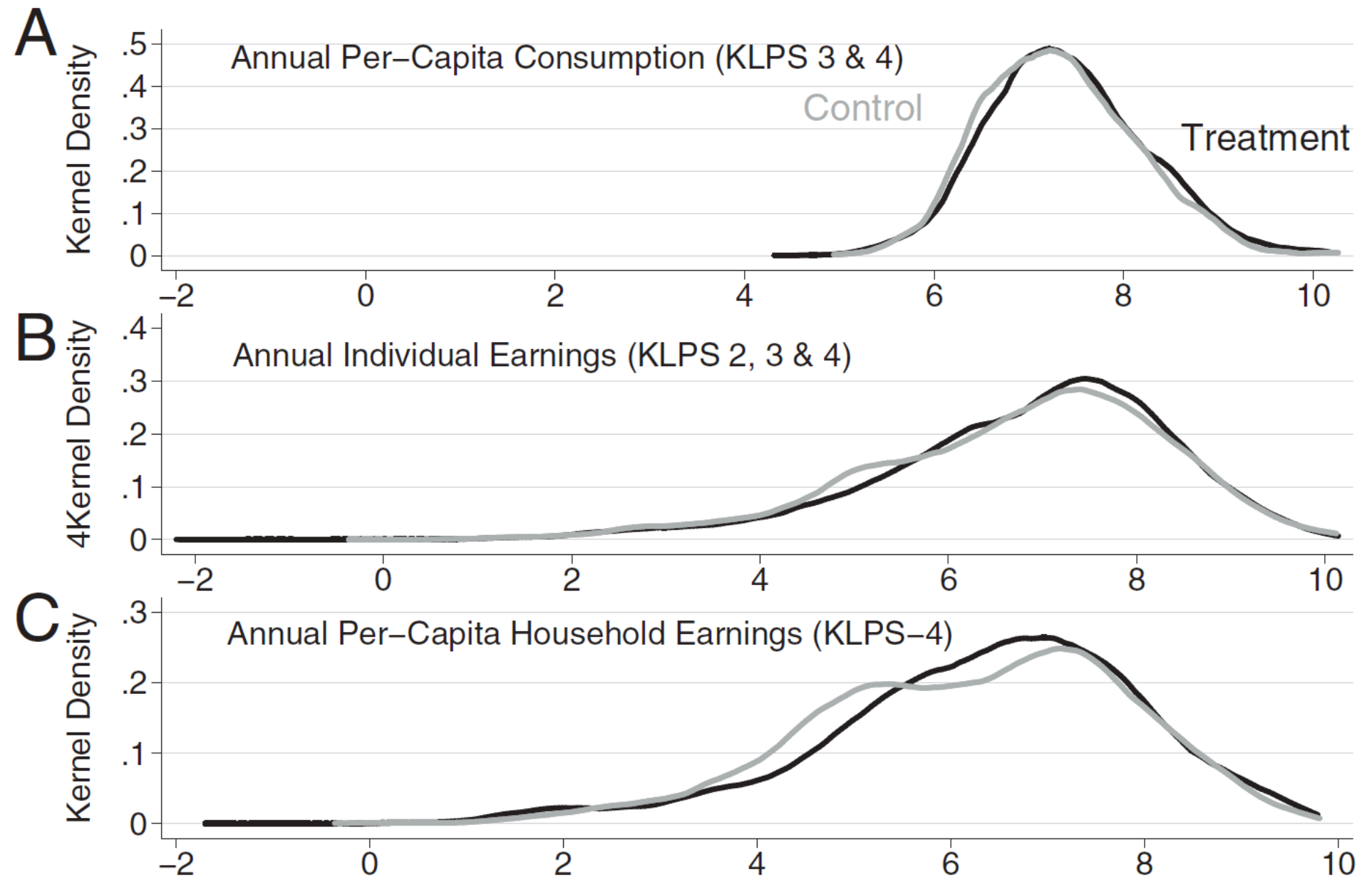
- Check the full table and interpret it.

	(1) Full sample	(2) Female	(3) Male	(4) Older	(5) Younger
A: Annual per capita consumption (KLPS-3 and KLPS-4)					
Treatment (λ_1)	305* (159)	89 (134)	513* (304)	886*** (223)	−179 (185)
Control mean	2,156	1,715	2,594	1,908	2,381
Treatment effect (%)	14.15	5.21	19.76	46.44	−7.52
Treatment <i>P</i> value	0.058	0.505	0.096	0.000	0.337
FDR <i>q</i> value	0.132	0.630	0.623	0.001	0.290
Number observations	4,794	2,473	2,321	2,402	2,341
B: Annual individual earnings (KLPS-2, KLPS-3, and KLPS-4)					
Treatment (λ_1)	80 (76)	41 (62)	118 (133)	258** (108)	−75 (100)
Control mean	1,218	674	1,728	1,177	1,242
Treatment effect (%)	6.53	6.02	6.84	21.93	−6.07
Treatment <i>P</i> value	0.297	0.515	0.376	0.019	0.451
FDR <i>q</i> value	0.175	0.630	0.630	0.030	0.292
Number of observations	13,624	6,826	6,798	6,791	6,780
C: Annual per capita household earnings (KLPS-4)					
Treatment (λ_1)	239* (129)	36 (107)	439* (252)	565** (232)	−22 (171)
Control mean	1,296	973	1,623	1,082	1,501
Treatment effect (%)	18.44	3.68	27.06	52.17	−1.48
Treatment <i>P</i> value	0.069	0.738	0.086	0.017	0.897
Number of observations	4,074	2,099	1,975	2,039	1,982

Long-term economic effects (2)

Source:
([Hamory et al. 2021](#))
Figure 1.

- the x-axis represents the log of annual earnings per capita.



Long-term economic effects (3)

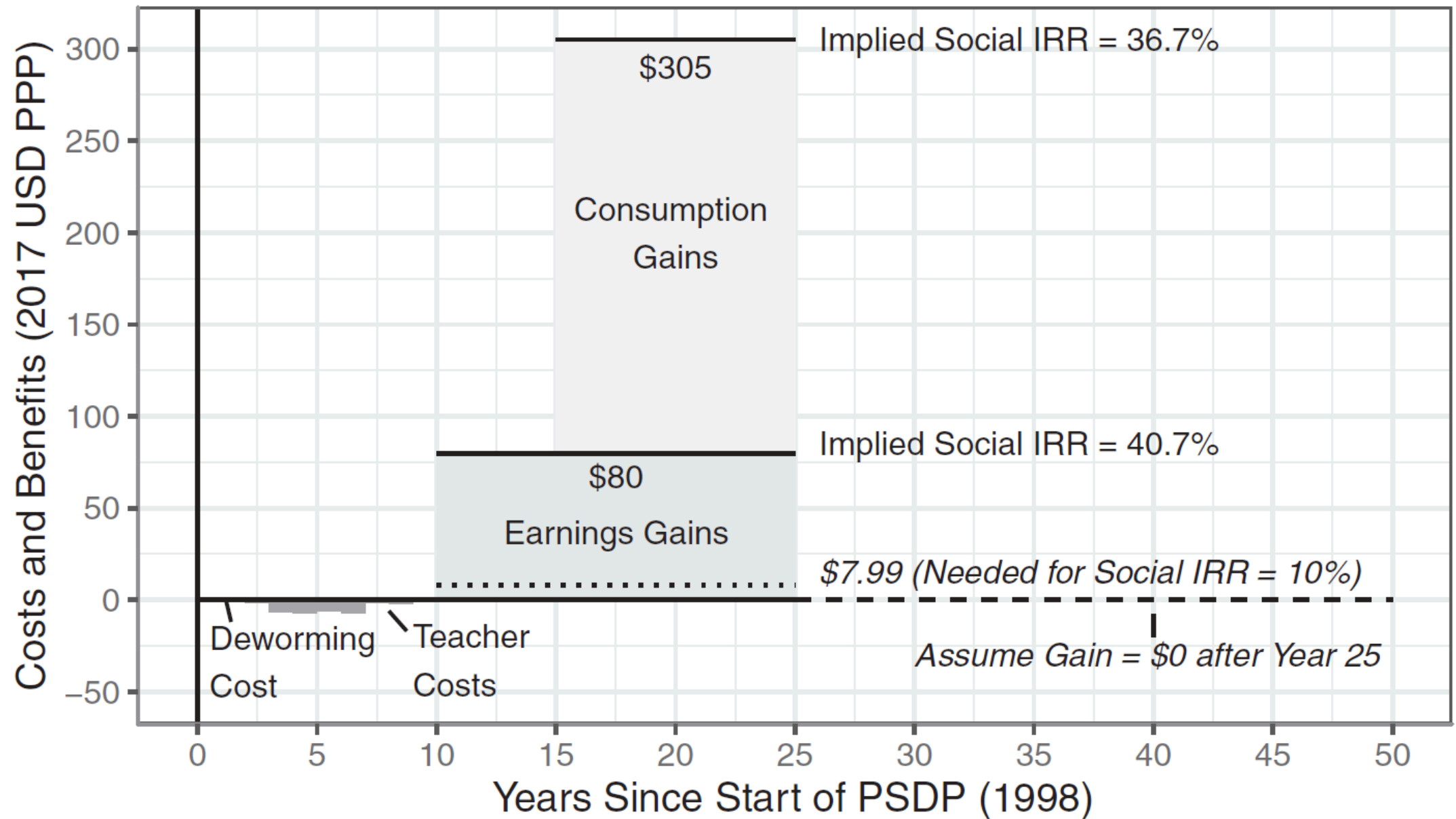
Source: ([Hamory et al. 2021](#)) Table 2.

Table 2. The 10- to 20-y deworming treatment effects on earnings, labor supply, occupation, and sectoral choice, KLPS-2, KLPS-3, and KLPS-4

	Treatment (λ_1)			Full sample	
	(1) Full sample	(2) Male	(3) Older	(4) Control mean	(5) Number of obs.
A: Earnings and wealth					
Log annual individual earnings	0.09 (0.06)	0.06 (0.07)	0.19** (0.08)	6.73	7,698
Wage earnings (annual)	81 (68)	138 (110)	162* (89)	887	13,628
Self-employment profit (annual)	41* (24)	51 (48)	70* (39)	212	13,638
Individual farming profit (annual)	−0 (2)	1 (3)	−3 (3)	9	13,707
Nonzero earnings	0.02* (0.01)	0.04** (0.02)	0.02 (0.02)	0.59	13,794
Hourly earnings	0.14* (0.08)	0.22 (0.15)	0.32* (0.16)	1.07	6,096
Per capita household wealth (KLPS-4)	69 (50)	102 (97)	253*** (89)	522	4,085
B: Labor supply, occupation, and sectoral choice					
Urban residence	0.04** (0.02)	0.06** (0.03)	0.03 (0.03)	0.45	13,793
Total hours worked (last 7 d)	1.04 (0.66)	2.20** (0.92)	1.79** (0.91)	24.19	13,807
Hours worked—agriculture (last 7 d)	−0.87** (0.43)	−0.57 (0.62)	−0.46 (0.56)	3.99	13,807
Hours worked—nonagriculture (last 7 d)	1.91*** (0.65)	2.77*** (0.94)	2.24** (1.08)	20.20	13,807
Employed—agriculture/fishing	−0.003 (0.008)	−0.001 (0.013)	0.004 (0.012)	0.043	13,768
Employed—services/wholesale/retail	0.002 (0.014)	0.012 (0.020)	−0.002 (0.019)	0.230	13,761
Employed—construction/trade contractor	0.004 (0.007)	0.011 (0.014)	−0.007 (0.009)	0.033	13,760
Employed—manufacturing	−0.001 (0.004)	0.002 (0.007)	0.002 (0.006)	0.026	13,760

Costs and benefits of deworming

Source:
([Hamory et al. 2021](#))
Figure 2



References

Example: Conditional Cash Transfers (CCTs)

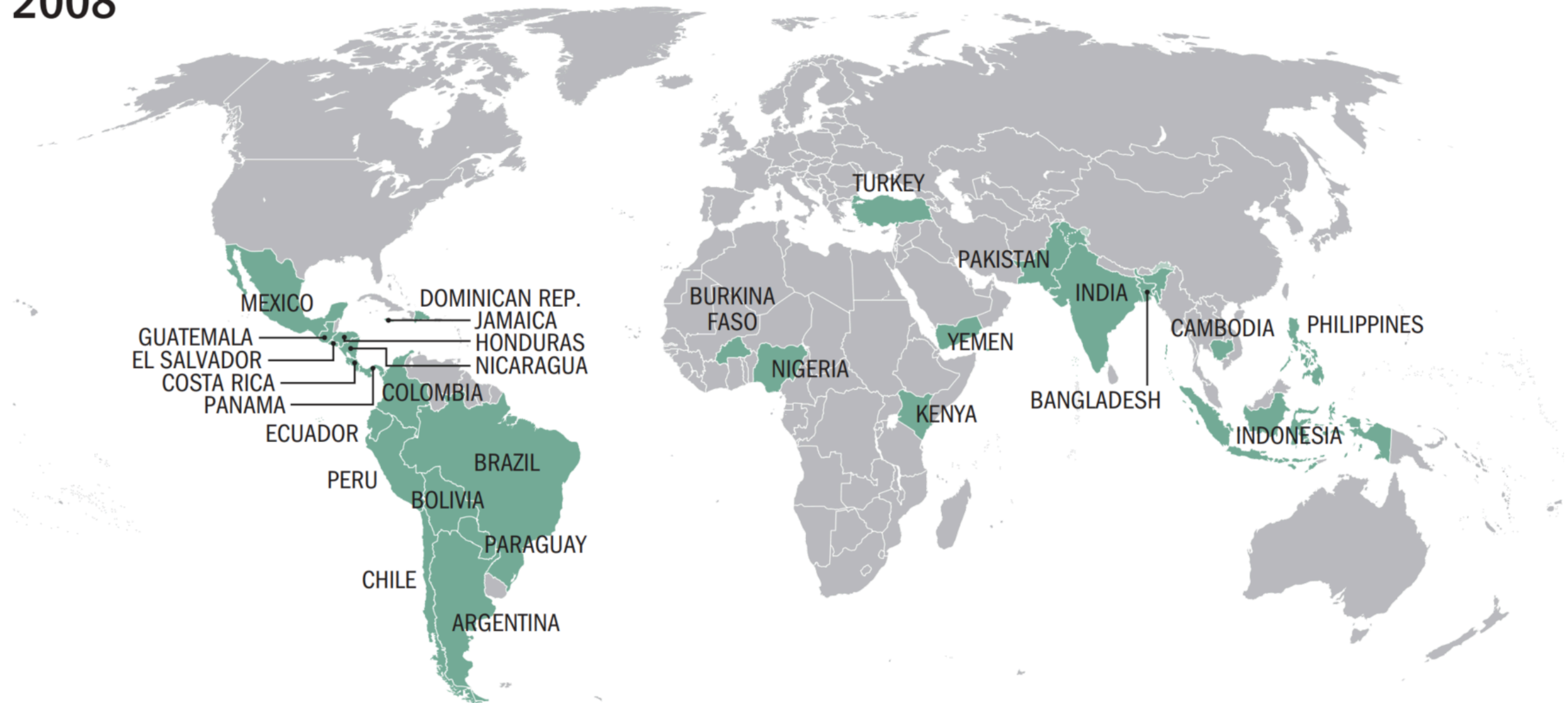
Let us consider an example of CONDITIONAL CASH TRANSFERS' causal effect on children's school enrollment.

See: CH 5. The Impact of CCT Programs on the Accumulation of Human Capital from ([Fiszbein et al. 2009](#)).

Conditional Cash Transfers (CCTs)

CCTs are programs that transfer cash on the condition of making pre-specified investments in human capital and children. ([Fiszbein et al. 2009, fig. 1](#))

2008



Cause and Effect

1. Cause / Treatment:

- CCTs (monetary incentive) + Pre-specified conditions

2.a Outcome / Factual

- School enrollment
- in treatment;

2.b Counterfactual

- School enrollment
- in control;

3. ATE: Effect = Factual – Counterfactual

- Average Treatment Effect = Enrollment Rate Difference

What makes “Difference in Enrollment Rate” a causal effect?

What statistical concepts ensure that the difference in school enrollment between the treatment and control is caused by the treatment?

1. Research Designed with the Randomized Control Trial (RCT):
 - Randomization between the Control and Treatment group;
2. **the Law of Large Numbers (LLN):**
 - Both random groups are equal on average;
3. The RCT + LLN = **Ceteris Paribus**;
 - All other things are equal between the two groups;

Good controls in the RCT

The control group is a perfect **counterfactual** when **three conditions are met**:

1. has the same characteristics, on average, as the treatment group in the absence of the program;
2. remains unaffected by the treatment; and
3. reacts to the treatment in the same way as the treatment group;

When these three conditions are met, the treatment causes the effect ceteris paribus ([P. J. Gertler et al. 2016](#)).

How to check that ceteris paribus is ensured?

What statistical tools are used to check that control and treatment groups are the same on average?

- Checks and balance tables
 - performs a mean-difference test between treatment and control for
 - variables that are (should be) independent of treatment
- No significant differences (failing to reject H_0) suggest that the two groups are the same on average
- Any significant differences (rejecting H_0 at the α level of significance) imply that the two groups are not random.

Ceteris Paribus

Other things are equal! The “other things” are:

- **measurable features** (age, height, costs, ...);
- **measurable but absent from our data;**
- **nonmeasurable factors** (ability, effort, connections, ...)

t-tests (1)

Helps us perform HT about the population based on (a) sample(s).

1. One sample t-test:

- is the population parameter μ different from an arbitrary number?

2. Two sample t-test (not covered):

- are two parameters μ_1 and μ_2 different in the population?

t-tests (2) One sample t-test algorithm

Hypothesis

$$H_0 : \mu = 0$$

$$H_1 : \mu \neq 0, \text{ or } H_1 : \mu > 0, \text{ or } H_1 : \mu < 0$$

Statistics

- Estimated parameter: $\hat{\mu}$
- Standard Deviation: s
- Zero hypothesis value: $\mu_0 = 0$
- Number of observations: n
- t-statistics: $t = \sqrt{n}(\hat{\mu} - \mu_0)/s$
- Significance level: $\alpha = 0.05$

Conclusions:

1. Reject H_0 if:

- $|t| > c_{df, 1-\alpha/2}$ or $|t| > c_{df, 1-\alpha}$, depending on the H_1 and
- p-value $< \alpha$ (probability of type I error $<$ level of significance)

2. Fail to reject H_0

t-tests (3.1) example

Do the CCTs improve children's school enrollment? ([Fiszbein et al. 2009](#))

- The effect: $\widehat{\text{Enrollment change}} = \widehat{\text{After the CCT}} - \widehat{\text{Before the CCT}}$
- Hypothesis:
 - H_0 : Enrollment change = 0
 - H_1 : Enrollment change > 0
- Statistical test:
 - One-sided t-test
- **Desired t-test result** at the 5% level of significance $\alpha = 0.05$.
 - Which t-test result indicates that CCT has a positive effect on school enrollment?
 - $\text{p-value} < \alpha$ or $\text{p-value} < 0.05$
 - $|\text{t-statistics}| > c_{df, 0.95}$
 - We need to be able to reject H_0

t-tests (3.2) example

Table 5.1 Impact of CCTs on School Enrollment and Attendance, Various Years

Country	Program	Age/Gender/Grade	Baseline enrollment (%)	Impact ^a	Transfer (% of PCE) ^b	Evaluation method	Reference
<i>Latin American and Caribbean countries</i>							
Chile	Chile Solidario	Ages 6–15	60.7	7.5*** (3.0)	7	RDD	Galasso (2006)
Colombia	Familias en Acción	Ages 8–13	91.7	2.1** (1.0)	17	PSM, DD	Attanasio, Fitzsimmons, and Gómez (2005)
		Ages 14–17	63.2	5.6*** (1.8)			
Ecuador	Bono de Desarrollo Humano	Ages 6–17	75.2	10.3** (4.8)	10	IV, randomized	Schady and Araujo (2008)
Honduras	Programa de Asignación Familiar	Ages 6–13	66.4	3.3*** (0.3)	9	Randomized	Glewwe and Olinto (2004)
Jamaica	Program of Advancement through Health and Education	Ages 7–17	18 days ^c	0.5** (0.2)	10	RDD	Levy and Ohls (2007)
Mexico	Oportunidades	Grades 0–5	94.0	1.9 (25.0)	20	Randomized	Schultz (2004)
		Grade 6	45.0	8.7*** (0.4)			
		Grades 7–9	42.5	0.6 (56.4)			
Nicaragua	Atención a Crisis	Ages 7–15	90.5	6.6*** (0.9)	18	Randomized	Macours and Vakis (2008)
Nicaragua	Red de Protección Social	Ages 7–13	72.0	12.8*** (4.3)	27	Randomized	Maluccio and Flores (2005)

- The column for “impact” reports the difference and standard error (in parentheses); the unit is percentage points, except of the Jamaican PATH program.
- *, **, and *** indicates if the p-value is below a corresponding significance levels of 10, 5, and 1 percent.
- What is shown in the rows?
 - case-specific t-test
- What are H_0 and H_1 ?
 - $H_0 : \text{diff} = 0; H_1 : \text{diff} \neq 0;$
- What are the conclusions for each row?
 - Reject/Accept H_0
- What do “stars” imply for test-specific conclusions for each row?

t-tests (3.3) example

Interpret results of the statistical test for country A at the X (1%, 5%) level of significance.

- State Null and Alternative hypotheses.
- Conclude about the HT results.
- What parameter/statistics/number was used to make this conclusion?

For example, in Turkey, Ahmed et al. (2007) failed to show that CCTs improve/worsen children's school enrollment. With "H0: no improvement" versus the "H1: some improvement", the authors failed to reject the H0 at the 5% sign. level for populations of primary and secondary school attendees ...

Table 5.1 continued

Country	Program	Age/Gender/Grade	Baseline enrollment (%)	Impact ^a	Transfer (% of PCE) ^b	Evaluation method	Reference
<i>Non-Latin American and Caribbean countries</i>							
Bangladesh	Female Secondary School Assistance Program	Ages 11–18 (girls)	44.1	12.0** (5.1)	0.6	FE	Khandker, Pitt, and Fuwa (2003)
Cambodia	Japan Fund for Poverty Reduction	Grades 7–9 (girls)	65.0	31.3*** (2.3)	2–3	DD	Filmer and Schady (2008)
Cambodia	Cambodia Education Sector Support Project	Grades 7–9	65.0	21.4*** (4.0)	2–3	RDD	Filmer and Schady (2009c)
Pakistan	Punjab Education Sector Reform Program	Ages 10–14 (girls)	29.0	11.1*** (3.8)	3	DDD	Chaudhury and Parajuli (2008)
Turkey	Social Risk Mitigation Project	Primary school	87.9	–3.0* n.a.	6	RDD	Ahmed et al. (2007)
		Secondary school	39.2	5.2 n.a.			

Source: Authors' compilation.
Note: DD = difference-in-differences; DDD = difference-in-difference-in-differences; FE = fixed effects; IV = instrumental variables; n.a. = not available; PCE = per capita expenditure; PSM = propensity score matching; RDD = regression discontinuity design. This table contains unweighted means for the coefficients for Colombia ages 8–13 and 14–17, Chile ages 4–5 and 6–15, and Mexico grades 0–5 and 7–9. The standard errors in each case are the square roots of the averaged variances of these estimates.
 a. The column for "impact" reports the coefficient and standard error (in parentheses); the unit is percentage points, with the exception of the Jamaican PATH program, where the unit is days.
 b. The transfer amounts as a proportion of per capita expenditures (or consumption) are not the same across all tables in the report because of differences in the surveys used, including their coverage and year.
 c. Impacts were measured in Jamaica only for student attendance over a 20-day reference period. The baseline enrollment rate prior to PATH was 96 percent.

(For example cont.) ... In absolute terms, authors show a reduction in primary school enrollment by 3% and an increase in secondary school enrollment by 5.2%. However, the probability that these changes are due to sample variation (probability of type I error / p-value) is above 5% and 10% respectively. Table source: (Fiszbein et al. 2009)

Homework: Read/do more on Hypothesis testing

[OpenIntro](#) Foundations for Inference:

- [Point Estimates](#)
- [Hypothesis Testing Fundamentals](#)
- [Inference for Estimators Other Than the Mean](#)

Khan Academy:

- [The idea of significance tests](#)

Homework

Make the topic-specific homework.

Answer sample exam questions.

Readings

- Chapter 1 Randomized Trials. In Mastering 'metrics : the path from cause to effect / Joshua D. Angrist and Jörn-Steffen Pischke. Princeton University Press, 2015. <https://justfind.hds.hebis.de/Record/HEB35797798X>
- Chapter 2 Regression. In Mastering 'metrics : the path from cause to effect / Joshua D. Angrist and Jörn-Steffen Pischke. Princeton University Press, 2015. <https://justfind.hds.hebis.de/Record/HEB35797798X>

Key concepts covered in the lecture

RCT:

- Causal meaning of the averages;
- Random sample variation;
- Need of the statistical hypothesis testing;

Standard Errors:

- Definition and importance.
- SE of the sample estimates of mean.
- Difference from the sample variance.

Hypothesis testing:

- Type I and Type II errors;
- Significance level;
- Null and Alternative Hypotheses;
- What is the probability value (p-value)?
- What do false-negative/false-positive mean?

References

- Carter, Susan Payne, Kyle Greenberg, and Michael S. Walker. 2017. “The Impact of Computer Usage on Academic Performance: Evidence from a Randomized Trial at the United States Military Academy.” *Economics of Education Review* 56 (February): 118–32. <https://doi.org/10.1016/j.econedurev.2016.12.005>.
- Fiszbein, Ariel, Norbert Schady, Francisco H. G. Ferreira, Margaret Grosh, Niall Keleher, Pedro Olinto, and Emmanuel Skoufias. 2009. *Conditional Cash Transfers : Reducing Present and Future Poverty*. The World Bank. <https://doi.org/10.1596/978-0-8213-7352-1>.
- Gertler, Paul. 2004. “Do Conditional Cash Transfers Improve Child Health? Evidence from PROGRESA’s Control Randomized Experiment.” *American Economic Review* 94 (2): 336–41. <https://doi.org/10.1257/0002828041302109>.
- Gertler, Paul J., Sebastian Martinez, Patrick Premand, Laura B. Rawlings, and Christel M. J. Vermeersch. 2016. *Impact Evaluation in Practice, Second Edition*. Washington, DC: Inter-American Development Bank; World Bank. <https://doi.org/10.1596/978-1-4648-0779-4>.
- Hamory, Joan, Edward Miguel, Michael Walker, Michael Kremer, and Sarah Baird. 2021. “Twenty-Year Economic Impacts of Deworming.” *Proceedings of the National Academy of Sciences* 118 (14). <https://doi.org/10.1073/pnas.2023185118>.
- Kuchenbecker, Judith, Anika Reinbott, Beatrice Mtimuni, Michael B. Krawinkel, and Irmgard Jordan. 2017. “Nutrition Education Improves Dietary Diversity of Children 6-23 Months at Community-Level: Results from a Cluster Randomized Controlled Trial in Malawi.” Edited by Jacobus P. van Wouwe. *PLOS ONE* 12 (4): e0175216. <https://doi.org/10.1371/journal.pone.0175216>.
- Miguel, Edward, and Michael Kremer. 2004. “Worms: Identifying Impacts on Education and Health in the Presence of Treatment Externalities.” *Econometrica* 72 (1): 159–217. <https://doi.org/10.1111/j.1468-0262.2004.00481.x>.

Neyman, Jerzy, and Egon Sharpe Pearson. 1933. "IX. On the Problem of the Most Efficient Tests of Statistical Hypotheses." *Philosophical Transactions of the Royal Society of London.*, 289–337.

Wooldridge, M. Jeffrey. 2020. *Introductory Econometrics: A Modern Approach*. South-Western.
<https://www.cengage.uk/shop/isbn/9781337558860>.